

HW06 — AI Audit Report

Student ID: 23127531
Assignment: HW06 — API Testing
AI tool: ChatGPT
Model: GPT-5.6 Sol
Primary usage dates: 2026-09-02 to 2026-09-03
Local timezone: UTC+07:00
Repository: <https://github.com/aohymevoL666/software-testing-homework6>

Timestamp note: The retained project transcript reliably preserves the interaction dates and order, but not an exact clock time for every individual AI exchange. Before final submission, exact times can be copied from the ChatGPT conversation UI if the instructor requires per-message clock timestamps. The audit below does not invent missing times.

AI usage policy

AI was used for test ideation, audit support, Postman/Newman artifact generation, execution-result interpretation, CI/CD scaffolding, spreadsheet consolidation, and documentation drafting. AI output was not accepted blindly: generated cases were compared with the supplied API specification and inspected SUT implementation, then classified as VALID / INVALID / INCOMPLETE before execution.

Interaction log

#	Date	Time	User prompt / interaction summary	AI output / assistance	Human review / action
1	2026-09-02	Not preserved	Establish HW06 scope, environment, and select one API from each pool.	Proposed and retained the baseline: POST /api/register, POST /api/cart, GET /api/admin/users; organized repository structure and execution plan.	Student accepted the three-API baseline and created the workspace.
2	2026-09-02	Not preserved	Inspect API 1 registration specification and implementation.	Identified request fields, response behavior, database schema, and missing validation constraints.	Student verified source and backend availability.

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3	2026-09-02	Not preserved	Generate registration API tests with broad partition/security/schema coverage.	Generated 42 AI test cases in api1_registration_ai_generated.md.	Student saved and committed the AI-generated artifact.
4	2026-09-02	Not preserved	Audit all AI-generated registration cases against the actual specification/implementation.	Classified 42 cases as 11 VALID, 5 INVALID, 26 INCOMPLETE; explained fabricated validation/uniqueness assumptions.	Student saved and committed the audit.
5	2026-09-02	Not preserved	Extend API 1 coverage after the audit.	Proposed six additional candidate cases covering Content-Type handling, null body, duplicate/case-variant email, and mass assignment.	Student reviewed, accepted, saved, and committed the extension.
6	2026-09-02	Not preserved	Implement and execute API 1 in Postman/Newman.	Generated Postman collection/environment, added collection-level X-Student-Id, interpreted Newman results, and drafted execution/bug artifacts.	Student executed Newman; two 500 responses were confirmed and evidence was captured.
7	2026-09-02	Not preserved	Inspect API 2 shopping-cart implementation.	Identified bearer authentication, in-memory	Student confirmed source and retained FR-07

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				userCarts, and absence of payload/prod uct validation.	as API 2.
8	2026-09-02	Not preserved	Generate and audit API 2 cases.	Generated 42 cases, then audited them as 13 VALID, 2 INVALID, 27 INCOMPLETE.	Student saved and committed both artifacts.
9	2026-09-02	Not preserved	Extend API 2 after AI audit.	Proposed six candidate state/security cases: metadata inconsistency, user isolation, stale JWT characterizati on, nested payloads, extra-field persistence, and restart persistence.	Student reviewed, accepted, saved, and committed the extension.
10	2026-09-02	Not preserved	Implement API 2 Postman collection and execute it.	Generated 62- request collection plus environment; automated two-user setup and 47 logical cases, leaving restart persistence for manual execution.	Student ran Newman: 109/109 assertions passed.
11	2026-09-03	Not preserved	Complete manual cart restart-state test.	Diagnosed an invalid first attempt caused by an empty email and backend reseeding; then proposed a seeded-user rerun.	Student reran using test@eshop.c om; cart contained marker before restart and [] after restart. Classified as

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					characterization, not a spec defect.
12	2026-09-03	Not preserved	Inspect API 3 Admin users route and authorization middleware.	Found that GET /api/admin/users uses authenticateToken only and performs no Admin role check; identified response fields.	Student accepted this as the execution baseline.
13	2026-09-03	Not preserved	Generate API 3 access-control tests.	Generated 42 AI cases covering auth, authorization, role claims, stale state, privilege escalation, response schema, and protocol handling.	Student saved and committed the generation artifact.
14	2026-09-03	Not preserved	Audit all API 3 AI cases.	Classified 26 VALID and 16 INCOMPLETE; separated explicit Admin-only requirements from unsupported token-revocation assumptions.	Student saved and committed the audit.
15	2026-09-03	Not preserved	Extend API 3 coverage.	Proposed six candidate cases emphasizing direct non-admin access, end-to-end	Student reviewed, accepted, saved, and committed the extension.

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				self-promotion, PII disclosure, wrong auth scheme, stale privilege, and response privacy.	
16	2026-09-03	Not preserved	Implement and execute API 3 in Postman/Newman.	Generated 47-request collection; retained strict FR-12 assertions rather than forcing a green run. Interpreted 18 failures into three genuine bugs plus one oracle correction.	Student executed Newman and captured evidence.
17	2026-09-03	Not preserved	Draft API 3 bug reports.	Produced Markdown reports for broken Admin access control, Bearer-scheme enforcement, and vertical privilege escalation.	Student committed reports and created GitHub Issues #2–#4.
18	2026-09-03	Not preserved	Publish repository and document confirmed bugs as GitHub Issues.	Provided issue titles/bodies and a linking artifact.	Student created four public GitHub Issues and committed docs/github_issues.md.
19	2026-09-03	Not preserved	Add CI/CD evidence with one passing and one intentional failing test.	Generated GitHub Actions workflow, deterministic CI smoke	Student produced green commit 170ab50, failing commit

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				collection/env ironment, and instructions for PASS – intentional FAIL – restore PASS.	31fa261, and restored passing state in 916e0ff.
20	2026-09-03	Not preserved	Capture anti- cheat X- Student-Id evidence in Postman Console.	Gave step-by- step GUI instructions, diagnosed {{baseurl}} vs {{baseUrl}}, and verified the final screenshot.	Student captured and committed evidence/post man_console_ student_id.pn g.
21	2026-09-03	Not preserved	Consolidate test cases and summary into the required spreadsheet artifact.	Generated HW06_Test_C ases_and_Su mmmary_2312 7531.xlsx with Summary, three API sheets, and Bug Register.	Student saved and committed workbook in 402ac63.
22	2026-09-03	Not preserved	Prepare mandatory AI audit and critique documentatio n.	Generated this audit and the separate 200– 300 word critique.	Student must review wording and fill exact clock times only if required by the instructor.

Important human-audit examples

Example 1 — API 1 validation assumptions

AI initially proposed expectations such as strict email-format validation, password rules, and duplicate-email rejection. Human/source review found that the supplied implementation and database schema did not define those constraints. Those cases were therefore marked INVALID or INCOMPLETE rather than being executed with fabricated oracles.

Example 2 — API 2 product/cart semantics

AI initially assumed that a nonexistent product id should return 404 and that repeated adds should merge quantities. Source review showed that POST /api/cart simply pushes the request body into an in-memory array. These assumptions were rejected during human audit.

Example 3 — API 3 authorization

The Admin-only rule was explicitly supported by the supplied API specification. Therefore normal-user denial cases remained VALID even though source inspection predicted they would fail. Execution later confirmed the defect: normal users received 200 OK and the global user list.

Disclosure about the “human extension” artifacts

AI assisted with proposing and formatting candidate extension cases after the audit. The student reviewed, selected, executed, and accepted the final cases. Because the assignment wording asks for student-added cases that AI missed, the safest academic-integrity approach is for the student to independently revise the wording or substitute at least five personally authored extension cases per API before final submission if the instructor interprets “of your own” as prohibiting AI-assisted ideation.

Files produced with AI assistance

- test-cases/api1_registration_ai_generated.md
- test-cases/api1_registration_audit.md
- test-cases/api1_registration_human_extension.md
- postman/HW06_API1_Registration.postman_collection.json
- docs/api1_registration_execution_summary.md
- reports/bugs/bug_api1_registration_content_type_500.md
- test-cases/api2_cart_ai_generated.md
- test-cases/api2_cart_audit.md
- test-cases/api2_cart_human_extension.md
- postman/HW06_API2_Cart.postman_collection.json
- docs/api2_cart_execution_summary.md
- test-cases/api3_access_control_ai_generated.md
- test-cases/api3_access_control_audit.md
- test-cases/api3_access_control_human_extension.md
- postman/HW06_API3_AccessControl.postman_collection.json
- docs/api3_access_control_execution_summary.md
- three API 3 security bug reports
- .github/workflows/api-tests.yml
- postman/HW06_CI_Smoke.postman_collection.json
- postman/HW06_CI.postman_environment.json
- docs/ci_cd_report.md
- test-cases/HW06_Test_Cases_and_Summary_23127531.xlsx

Human responsibility statement

The student remained responsible for running commands, inspecting the SUT, executing Newman/Postman, collecting screenshots, reviewing AI-generated cases, deciding which observed behaviors were genuine specification defects, publishing GitHub Issues, and committing/pushing the final artifacts.